W-A-450/8
JANUARY 3, 1992

FEDERAL SPECIFICATION

DETECTOR, PROXIMITY/CAPACITANCE

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION.

1.1 Scope. The W-A-450C specification covers intrusion detection equipment which is deployed as a system to include, but is not to be limited to, detection of intrusions relating to protection of national security information, sensitive information and equipment, high dollar items, and weapons. The intrusion detection equipment, when selectively assembled, provides interior detection capabilities specifically designed for security application.

1.2 Classification.

1.2.1 Unit and type. This specification covers Unit Number 1, Type VIII, Proximity/Capacitance Detector.

2. APPLICABLE DOCUMENTS.

The documents listed in W-A-450C/GEN, concerning the issues in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

3. REQUIREMENTS.

3.1 Qualification. The capacitance detector furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list. It shall conform to the minimum specifications required herein and shall have been tested and shall have passed the qualification tests specified in paragraph 4.1.

3.2 Temperature and humidity ranges.

3.2.1 Operating range. All equipment shall be capable of operating within the temperature and humidity ranges indicated below:

0 degrees C to +49 degrees C and relative humidity of 0% to 85%, non-condensing.

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All units shall withstand the tests in paragraphs 4.1.3 through 4.1.5. Additional requirements may be imposed by the procuring entity.

3.2.2 Nonoperating range. All equipment which is not powered up shall be capable of withstanding, without damage, the temperature range indicated below:

-46 degrees C to 71 degrees C.

3.3 Dependability. The capacitance detector supplied under this specification shall have a minimum Mean Time Between Failure (MTBF) of 10 000 hours.

3.4 Operation. The capacitance detector shall consist of a protection loop wire, necessary control circuitry, and any additional items such as isolation material and mounting hardware as is deemed necessary to adequately install an intrusion detection system. The capacitance detector shall, after initial setup, stabilize within five minutes after being powered on and after any alarm initiating stimulus is removed. A test target of a capacitance representative of a human shall be used to test the detector (see 4.1.1). The detector shall detect changes in capacitance associated with intrusion attempts. Input voltage shall be 8 to 18 volts DC, reverse polarity protected, or 10 to 28 volts AC RMS, 60 cps. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Single-pole, double-throw alarm switch or relay and separate single-pole, double-throw tamper switch or relay connections shall be provided at a terminal strip for connection to external alarm reporting equipment. Alarm and tamper switches shall provide both normally open and normally closed contacts. Switch impedance shall be as follows:

Open state:
- greater than 100 000 ohms.

Closed state:
- For solid state relays - less than 50 ohms.
- For mechanical relays - 0.1 ohms or less.

3.4.1 Additional capabilities. The manufacturer is encouraged to provide information in their technical literature concerning performance data that would exceed the minimum requirements of this specification.

3.5 Capacitance detector enclosure. The capacitance detector assembly and any tamper susceptible components shall be housed in enclosures having a removable cover arranged to provide ready access for mounting, wiring, alignment, and for sensitivity and other internal adjustments. Mounting holes shall be provided in the back of the enclosure in positions that are readily accessible (sufficiently displaced from all internal parts and the enclosure side) when all internal parts and wiring are in place. No external hardware or fasteners except the cover fasteners shall be used.

3.6 Sensitivity adjustment. While the capacitance detector is not required to have an adjustable sensitivity, this feature is highly desired and should allow for adjustments in the maximum capacitance change required for an alarm condition. If a sensitivity adjustment is provided, the detector shall detect throughout the entire sensitivity range.

3.7 Cover. The cover shall be designed such that any attempted access to the interior shall cause a tamper condition. The tamper circuits shall be in their secure state whenever the cover is in place and shall go to their alarm state when the cover is moved sufficiently to reveal the inside or to introduce a defeating device. The tamper switch shall be constructed of corrosion resistant materials and operate smoothly to prevent jamming.

3.8 Terminals. Screw terminals shall be provided for interconnecting wiring. Access to the terminals shall require removal of the enclosure cover.

3.9 Switch assembly and/or relay assembly. Alarm and tamper switch and/or relay contacts shall be protected against environmental and excessive current conditions which may be damaging. Sustained currents up to 1.33 times the listed or published current rating shall not cause damage to the switch or contacts.

3.10 Size, weight and power consumption. The size, weight and power consumption of each unit shall be the minimum practicable, consistent with meeting the requirements of this specification and shall be stated in the manufacturer's literature.

3.11. Maintainability. The capacitance unit shall be designed and constructed in such a manner as to minimize maintenance and repair time. To the maximum extent practicable, items shall be of a modular, unitized construction. Components shall be so arranged and assembled that they are readily visible and accessible to maintenance personnel. Specialized training is not required to maintain this equipment. Field maintenance shall consist of trouble shooting and isolation of failures to the device or replaceable sub-assembly, and replacement of the defective device or sub-assembly. Preventive maintenance may be required. Adjustments shall require no disassembly of the detector beyond mere removal of the cover.

3.12 Life cycle. The capacitance detector alarm and tamper switches and/or relay contacts shall be rated at not fewer than 1 000 000 operations under a load of 250 milliamperes driven from a 48 VDC source. Contact resistance shall not change more than ±20% from the initial reading.

3.13 EMI compatibility. Units shall operate normally when subjected to the testing of 4.1.6. The testing shall neither inhibit valid alarms nor generate invalid alarms.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Tests.

4.1.1 Performance test. The unit being tested shall be mounted, wired, and aligned in accordance with the manufacturer's recommendations to a standard test object. The standard test object is a 910 mm. square, 0.635 mm thick aluminum plate mounted in electrical isolation and centered 101.6 mm above a second, electrically grounded 1370 mm square, 0.635 mm thick aluminum plate (see figure 1).

4.1.1.1 Detection test. The unit being tested shall produce an alarm when a grounded 100 mm diameter aluminum sphere approaches the standard test object within the range of greater than 0 up to 100 mm at velocities that range from 0.3 m/s to 1.5 m/s. If unit being tested incorporates a sensitivity adjustment, this test shall be performed at both full and minimum sensitivities.

4.1.1.2 Capacitance load test. The unit being tested shall be tested in accordance with 4.1.1.1 above but with the addition of loading capacitors so that the total capacitive load presented to the detector is equal to the total load specified by the manufacturer as maximum.

4.1.1.3 Recovery time. The units submitted for qualification testing shall, after initial setup, stabilize within five minutes after being powered on and within 2 minutes after any alarm stimulus is removed. During the performance of Detection Test (see 4.1.1.1) the recovery time shall be measured after each power on and post alarm condition. A unit having a recovery time in excess of the above times shall be disqualified.

4.1.2 Stability test. The unit submitted for qualification testing shall perform its intended function in a stable environment free of stimuli that would initiate a nuisance alarm. The test period shall be 90 days of continuous operation with any adjusted controls set to provide the specified security. The test period shall commence after the unit has been installed in accordance with the manufacturer's instructions and after a "shakedown" operational period of not more than seven consecutive days.

4.1.3 High temperature test. Units submitted for qualification shall be energized for a period of four hours at 49 degrees C. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 49 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

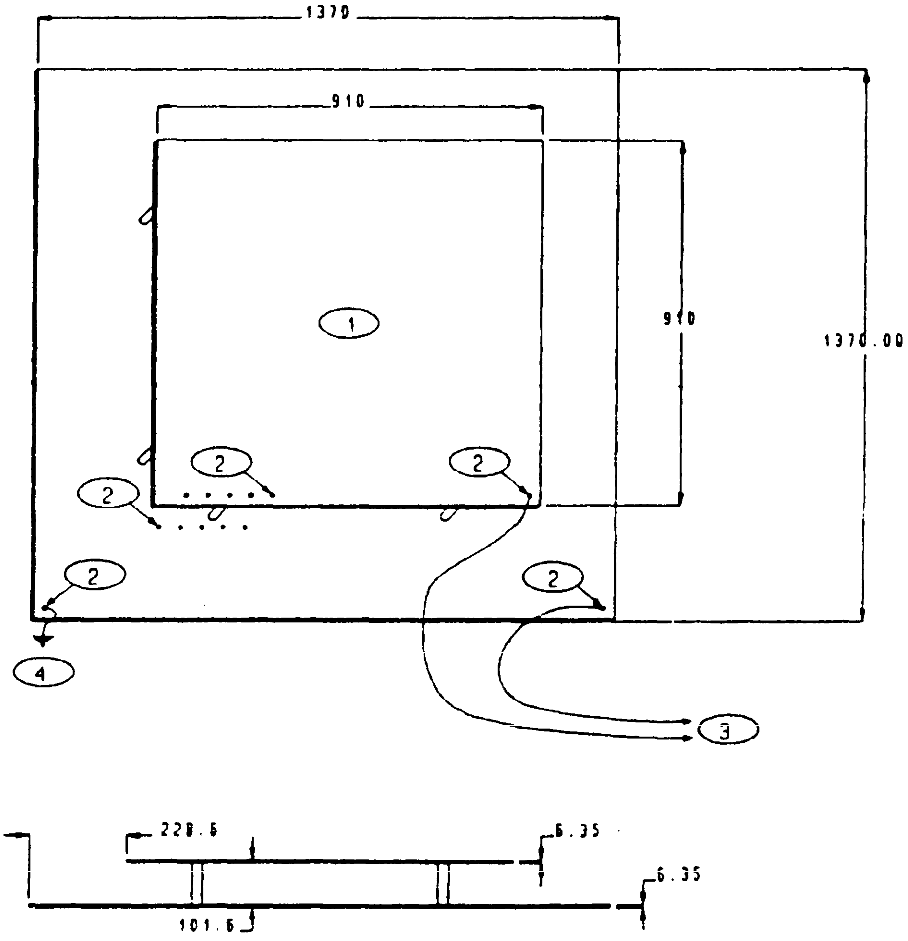
4.1.4 Low temperature test. Units submitted for qualification shall be energized for a period of four hours at 0 degrees C. The unit shall then be tested as specified in paragraph 4.1.1 with the temperature maintained at 0 degrees C. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature must be stabilized for 5 minutes before the test period commences.

4.1.5 Humidity test. Units submitted for qualification shall be energized for a period of four hours with the humidity maintained at 85% at a maintained temperature of at least 26 degrees C. The test chamber shall be vented to the atmosphere to prevent the accumulation of moisture. The unit shall be tested as specified in paragraph 4.1.1 with the temperature maintained at 26 degrees C and the humidity maintained at 85%. During this period, there shall be no alarms other than those resulting from the performance of the tests of 4.1.1. The temperature and humidity must be stabilized for 5 minutes before the test period commences.

4.1.6 EMI compatibility test. Units shall be tested for compliance with 3.13 in accordance with the applicable tests of MIL-STD-462 with consideration of the following paragraphs:

4.1.6.1 Test chamber. All testing shall be performed in an EMI test chamber.

4.1.6.2 Conducted electromagnetic susceptibility test. The units shall be subjected to power line conducted interference in accordance with test methods CS01, CS02, and CS06 of MIL-STD-462. Maintain a test level of 1 or 3 Volts RMS as required over the entire frequency range.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NOTES:

① UPPER PLATE TO BE CENTERED OVER LOWER PLATE

② DRILL AND TAP FOR 4 OR 6 MM. SCREW

③ WIRE CONNECTIONS TO ALARM PROCESSOR

④ LOWER PLATE TO BE CONNECTED TO GOOD EARTH GROUND

Figure 1
Standard Test Object

(a) more than 3/4 inch

4.1.6.3 Radiated electromagnetic susceptibility test. The units shall be subjected to radiation in accordance with test method RS03 of MIL-STD-462. Use an electric field of 1 Volt per meter for frequencies in the range of 14 Khz. to 2 Mhz. and 3 Volts per meter for frequencies in the range of 2 Mhz. to 12 Ghz.

5. PREPARATION FOR DELIVERY.

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing and marking shall be in accordance with ASTM D3951. The level of packing shall be A, B, or C, as specified (see 6.2).

5.1.1 Civil agency marking. In addition to any special markings (see 6.2) required by the contract or order, the interior packages and shipping container shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. Sensor systems constructed with components qualified under this specification are intended to supplement or support protective personnel (e.g., guards) for areas requiring special security measures. These properly assembled systems are resistive to neutralization or to compromise by covert or surreptitious attack. Guidance in selecting a particular item for a particular security situation is not intended and is beyond the scope of this document. Qualification approval merely means that the item meets the specification requirements at the time it was tested for qualification.

6.2 Ordering data. The purchaser should exercise any desired options permitted herein and include the following information in procurement documents:

(a) Title, symbol, and date of this specification.
(b) Type of coverage (detection pattern) required (where applicable).
(c) Quantities required.
(d) Spare parts required.
(e) Destination inspection required.
(f) Levels of packaging and packing required (see 5.1).
(g) Special marking required (see 5.2).
(h) Shipping schedule required.

6.3 Qualification. Regarding products which require qualification under this specification, awards will be made only for products which have been tested and approved for inclusion on the applicable Federal Qualified Products List prior to the time set for opening of bids whether or not such products have actually been listed. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products they propose to offer the Federal Government tested for qualification so that they may be eligible to be awarded contracts or orders for products covered by this specification. The activity responsible for the Qualified Products List is:

General Services Administration
Federal Supply Service, Furniture Commodity Center
Washington, DC. 20406

Information pertaining to qualification may be obtained from that activity.

6.4 Definitions.

6.4.1 Alarms.

6.4.1.1 Valid alarm. A valid alarm is any alarm generated by stimuli which the detector is designed to recognize and which satisfies the detector's stated alarm criteria.

6.4.1.2 Invalid alarm. Any other alarm.